# GEORGIOS CHRISTOU

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## EDUCATION

**PhD Particle Physics**, The University of Edinburgh, Edinburgh, Scotland — Sept 2024 - Present

**MSc Particle and Nuclear Physics**, The University of Edinburgh, Edinburgh, Scotland — Sept 2023 - Aug 2024
Graduated with A3 Distinction, $1^{st}$ in class, GPA: 75/100

**BSc Physics**, University of Cyprus, Nicosia, Cyprus — Sept 2019 - Jun 2023
Graduated with Excellence, $1^{st}$ in class, GPA: 8.66/10

## EXPERIENCE

**PhD Project**, The University of Edinburgh, United Kingdom — Sept 2024 - Present

*Analysis*

- Developed deep neural network classifiers for rare-event detection, lifting sensitivity by up to 25% over baseline.
- Added systematics to the statistical model accounting for event migration between categories.
- Improved a statistical model component, cutting mismodelling by 50% and deriving uncertainties for the residual.

*Simulation*

- Validated a fast approximate simulator against the full physics-based reference, tuning it per detector region.
- Built normalizing-flow generative models for detector response: within 5% of the reference at over 6x speedup.

**MSc Thesis**, The University of Edinburgh, United Kingdom — Nov 2023 - Aug 2024

- Benchmarked gradient-boosted trees against deep neural networks for multi-class classification on large simulated datasets, selecting the production model on cross-validated performance.
- Fed model outputs into a likelihood-based inference pipeline, delivering the first constraints on a set of physical parameters via a newly proposed method.

**CERN Summer Student Programme 2023**, Switzerland — Jun 2023 - Aug 2023

- Designed and implemented a C++ re-weighting algorithm correcting sampling bias between datasets, improving measurement precision over the incumbent method.
- Benchmarked against the previous approach on simulated data; report and code published publicly.

**BSc Thesis and Undergaduate Internship**, University of Cyprus, Cyprus — May 2022 - May 2023

- Analysed large-scale simulation output, applying multi-state fitting and model averaging to extract physical parameters with quantified uncertainty.
- Co-authored a peer-reviewed publication in Physical Review D reporting the first calculation of its kind.

## AWARDS & ACHIEVEMENTS

**Class Medal Award for MSc in Particle and Nuclear Physics**, The University of Edinburgh — Nov 2024

**Valedictorian in the Department of Physics**, University of Cyprus — Jun 2023

## SKILLS

- **Programming**: Python, C++, Bash/Shell, Fortran, Mathematica
- **Machine Learning**: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost; deep neural networks, generative models (normalizing flows), hyperparameter optimisation
- **Statistics & Data**: Hypothesis testing, maximum-likelihood inference, uncertainty quantification; NumPy, Pandas, Matplotlib, Seaborn
- **Scale & Tooling**: HPC & batch computing, columnar data at scale (ROOT/uproot, HDF5, Parquet/Arrow), Git, Linux/Unix, LaTeX
- **Languages**: Greek (Native), English (IELTS Score: 8, Level: C1), French (Beginner)